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SYSTEM AND METHOD FOR MANAGING SMART BUILDING FOR INTEGRATED OPERATION OF HETEROGENEOUS MOBILITY DEVICES

Abstract

A system comprises a smart building with a vertiport zone and a complex zone; and a managing device. The managing device is configured to: establish movement route information, of the aerial mobility device in the vertiport zone, based on management information associated with a transport object or flight of the aerial mobility device. If the movement route information indicates to move to at least one designated area of the vertiport zone and a charging permission condition is satisfied, the movement route information is set to add a charging site as a layover point. The aerial mobility device is controlled to move to the charging site according to the set movement route information and to be charged. If the movement route information indicates to move to the complex zone, the aerial mobility device is controlled to move to an exit area of the vertiport zone approaching the complex zone.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to Korean provisional patent application No. 10-2024-0033329, filed Mar. 8, 2024, the entire contents of which are incorporated herein for all purposes by this reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a system and method for managing a smart building for integrated operation of heterogeneous mobility devices, and for example, to a system and method for managing a smart building to improve operation processing of an aerial mobility device and enables the aerial mobility device to establish smooth support and an integrated link to a ground mobility device.

BACKGROUND

[0003] With advances in technology, various transportation mobility devices are emerging. To improve traffic congestion on the ground and urban environment deterioration, advanced aerial mobility (AAM) devices are being actively developed. An AAM device is capable of moving to a specific point in a complicated urban area and providing a service linked to another type of mobility device such as a ground mobility device.

[0004] A facility in which services can be provided between an AAM device and a ground mobility device (such a facility being referred to as a smart building or smart vertiport) is needed.

SUMMARY

[0005] The following summary presents a simplified summary of certain features. The summary is not an extensive overview and is not intended to identify key or critical elements.

[0006] Systems, apparatuses, and methods are described for managing a smart building. A system may comprise a smart building comprising: a vertiport zone, and a complex zone configured to support operation of an aerial mobility device, that lands in the vertiport zone, and a ground mobility device; and a managing device comprising: at least one processor, and a memory storing at least one instruction that, when executed by the at least one processor, configures the at least one processor to: generate, based on management information, movement route information of the aerial mobility device in the vertiport zone, wherein the management information is associated with management of at least one of: a transport object of the aerial mobility device, or flight of the aerial mobility device; based on the movement route information indicating at least one designated area of the vertiport zone and on a charging permission condition being satisfied, set the movement route information to comprise a layover point of a charging site in the vertiport zone and control the aerial mobility device to move to the charging site according to the set movement route information and to be charged, and based on the movement route information indicating to move the aerial mobility device to the complex zone, control the aerial mobility device to move to an exit area, of the vertiport zone, approaching the complex zone.

[0007] Also, or alternatively, a method may be for managing a smart building comprising a vertiport zone, and a complex zone configured to support operation of an aerial mobility device and

a ground mobility device. The method may comprise: generating, based on management information, movement route information of the aerial mobility device in the vertiport zone, wherein the management information is associated with management of at least one of: a transport object of the aerial mobility device, or flight of the aerial mobility device; based on the movement route information indicating at least one designated area of the vertiport zone and on a charging permission condition being satisfied, setting the movement route information to comprise a layover point of a charging site in the vertiport zone and controlling the aerial mobility device to move to the charging site according to the set movement route information and to be charged; and based on the movement route information indicating to move the aerial mobility device to the complex zone, controlling the aerial mobility device to move to an exit area, of the vertiport zone, approaching the complex zone.

[0008] These and other features and advantages are described in greater detail below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a view exemplifying a smart building according to the present disclosure.

[0010] FIG. 2 is a view exemplifying a vertiport zone.

[0011] FIG. 3 is a view showing constituent modules of a managing device according to an example of the present disclosure.

[0012] FIG. 4 is a view showing constituent modules of a mobility device according to the present disclosure.

[0013] FIG. 5 is a flowchart showing a method for managing a smart building for integrated operation of heterogeneous mobility devices according to another example of the present disclosure.

[0014] FIG. 6 is a flowchart showing a method for managing a smart building according to still another example of the present disclosure.

[0015] FIG. 7 is a flowchart showing a method for managing a smart building according to yet another example of the present disclosure.

DETAILED DESCRIPTION OF DISCLOSURE

[0016] Hereinafter, examples of the present disclosure are described in detail with reference to the accompanying drawings so that those having ordinary skill in the art may easily implement the present disclosure. However, examples of the present disclosure may be implemented in various different ways and thus the present disclosure is not limited to the examples described therein.

[0017] In describing examples of the present disclosure, well-known functions or constructions have not been described in detail since a detailed description thereof may have unnecessarily obscured the gist of the present disclosure. The same constituent elements in the drawings are denoted by the same reference numerals and a repeated or duplicative description of the same elements has been omitted.

[0018] In the present disclosure, if an element is simply referred to as being “connected to”, “coupled to” or “linked to” another element, this may mean that an element is “directly connected to”, “directly coupled to”, or “directly linked to” another element or this may mean that an element is connected to, coupled to, or linked to another element with another element intervening therebetween. In addition, If an element “includes” or “has” another element, this means that one element may further include another element without excluding another component unless specifically stated otherwise.

[0019] In the present disclosure, the terms first, second, etc. are only used to distinguish one element from another and do not limit the order or the degree of importance between the elements unless specifically stated otherwise. Accordingly, a first element in an example could be termed a

second element in another example, and, similarly, a second element in an example could be termed a first element in another example, without departing from the scope of the present disclosure.

[0020] In the present disclosure, elements are distinguished from each other for clearly describing each feature, but this does not necessarily mean that the elements are separated. In other words, a plurality of elements may be integrated in one hardware or software unit, or one element may be distributed and formed in a plurality of hardware or software units. Therefore, even if not mentioned otherwise, such integrated or distributed examples are included in the scope of the present disclosure.

[0021] In the present disclosure, elements described in various examples do not necessarily mean essential elements, and some of them may be optional elements. Therefore, an example composed of a subset of elements described in an example is also included in the scope of the present disclosure. Examples including other elements in addition to the elements described in the various examples are also included in the scope of the present disclosure.

[0022] The advantages and features of the present disclosure and the ways of attaining them should become apparent to those of ordinary skill in the art with reference to examples of the present disclosure described below in detail in conjunction with the accompanying drawings. The examples of the present disclosure, however, may be embodied in many different forms and should not be constructed as being limited to the example examples set forth herein. Rather, the examples described herein are provided to make this disclosure more complete and to fully convey the scope of the present disclosure to those having ordinary skill in the art to which the present disclosure pertains.

[0023] In the present disclosure, each of phrases such as “A or B”, “at least one of A and B”, “at least one of A or B”, “A, B or C”, “at least one of A, B and C”, and each of the phrases such as “at least one of A, B or C” and “at least one of A, B, C or combination thereof” may include any one or all possible combinations of the items listed together in the corresponding one of the phrases.

[0024] In the present disclosure, expressions of location relations used in the present specification such as “upper”, “lower”, “left” and “right” are employed for the convenience of explanation, and If drawings illustrated in the present specification are inversed, the location relations described in the specification may be inversely understood. If a component, device, element, or the like of the present disclosure is described as having a purpose or performing an operation, function, or the like, the component, device, or element should be considered herein as being “configured to” meet that purpose or perform that operation or function.

[0025] A smart building may be configured to process various tasks, such as fusion/interfaces between heterogeneous mobility devices (e.g., between an aerial mobility and a robot operated in the building), passenger transit, transshipment, and/or a link to a means of transportation between the building and a region outside the building.

[0026] Hereinafter, referring to FIG. 1 and FIG. 2, a smart building according to the present disclosure will be described. FIG. 1 is a view exemplifying a smart building according to the present disclosure. FIG. 2 is a view exemplifying a vertiport zone.

[0027] A smart building **100** may control aerial mobility devices **600** and/or **610** to carry transport objects through takeoff and landing. The smart building **100** may control the aerial mobility devices **600** and/or **610** to provide thereto a linked service of another mobility device (not shown) that moves outside the building. Also, or alternatively, the smart building **100** may be equipped with ground mobility devices **620** and **630** that support operation of the aerial mobility device **600** processed in the building. The ground mobility devices **620** and **630** may transfer a transport object of the aerial mobility devices **600** and **610** within the building or support energy supply and maintenance for the aerial mobility devices **600** and **610**, while traveling a plurality of zones and areas of the smart building **100**. Another mobility device, which carries a transport object to a building or the outside thereof, may include, for example, a conventional vehicle, a route vehicle, a

shared vehicle, a small personal mobility (e.g., electronic scooter, bicycle), and public transportation. Also, or alternatively, apart from a mobility service, the smart building **100** may have amenities for people who use the building.

[0028] For example, a transport object may be passengers and cargo. For example, the aerial mobility devices **600** and **610** may be, either or respectively, an AAM device capable of carrying a passenger and cargo and providing a personalized aerial service within a predetermined region, a small unmanned aerial vehicle (UAV or drone), which usually carries cargo or carries out a predetermined task such as photography and fluid jetting, a helicopter, and the like. For example, the ground mobility devices **620** and **630**, which move inside the building **100**, may include a robot and a purpose-based carrier (e.g., that may be implemented by autonomous driving/control).

[0029] Specifically, the smart building **100** may include a vertiport zone **200**, a complex zone **300**, and a supplementary zone **400**. In FIG. **1**, the zones **200**, **300** and **400** are exemplified in vertical arrangement but are not limited thereto and may be placed in various modes as long as the zones are separated from each other with respect to functions. Each of the zones may have a facility capable of moving between them, and the ground mobility devices **620** and **630** may move across the zones. The vertiport zone **200** may provide a take-off/landing space of the aerial mobility devices **600** and **610**, a waiting space of the devices, and a simple service of the devices and also have a board/alight space for a transport object. The complex zone **300** may be a zone where the aerial mobility devices **600** and **610**, which are going to take off in a predetermined time or require a specific work, are transported from the vertiport zone **200** and are kept. The supplementary zone **400** may be a zone which transport objects access to be carried by the aerial mobility devices **600** and **610**. For example, the supplementary zone **400** may have a space for holding another mobility device that provides a linked service of the aerial mobility devices **600** and **610** to enable the devices to be used.

[0030] Specifically, the vertiport zone **200** may include a passenger terminal **202** with a roof **204** installed thereon, a first take-off/landing area **206**, a taxiway **208**, an object board/alight area **210**, a rotator **212**, a charging site **214**, a waiting area **216**, a second take-off/landing area **218**, a storage area **220**, and a shift area **222**.

[0031] The passenger terminal **202** is an area for waiting to load passengers and cargo or unload them, and the cargo may be held or arranged by being distinguished according to the aerial mobility devices **600** and **610** where the cargo is loaded or unloaded. The roof **204** may be installed to prevent a transport object from being damaged in bad weather.

[0032] The first take-off/landing area **206** and/or the second take-off/landing area **218** may each be a site where the aerial mobility devices **600** and/or **610** take off and/or land. The first take-off/landing area **206** and/or the second take-off/landing area **218** may be provided only to the aerial mobility devices **600** and/or **610** with a specific type. For example, the aerial mobility device **600** may be an AAM device, which may use the first take-off/landing area **206**. The aerial mobility device **610** may be a UAV, which may be smaller than the AAM device, and may use the second take-off/landing area **218**. In FIG. **1**, the first take-off/landing area **206** and the second take-off/landing area **218** are each exemplified in a number of one but may be provided in a plural number. FIG. **1** exemplifies a take-off/landing area for both take-off and landing, but another example may separate a take-off area and a landing area. The AAM device may be equipped with wheels and thus move to a zone or area of the smart building **100** on its own. The UAV may not be equipped with any wheel and thus be moved manually by a cart that transports a device. Even if the AAM device is equipped with wheels, a cart may load and transport the AAM device.

[0033] The first take-off/landing area **206** and/or the second take-off/landing area **218** may be equipped with three separate sections. For example, the first take-off/landing area **206** and the second take-off/landing area **218** may include a touchdown and liftoff (TLOF) section, a final approach and takeoff (FATO) section, and a safety section. The TLOF may be a generally-paved area that supports take-off/landing loads of the aerial mobility devices **600** and/or **610**. The FATO

may be an area in a space where the aerial mobility devices **600** and/or **610** complete a final approaching step for hovering or landing, or an area in a space where the aerial mobility devices **600** and/or **610** start an initial step of taking off from a hovering or landing position. The TLOF may be at the center of the FATO. The safety area may be an area in a space that surrounds the FATO to provide an additional buffer zone.

[0034] The first take-off/landing area **206** and/or the second take-off/landing area **218** may be equipped with various sensing modules (e.g., sensors, such as weight sensors, timing sensors, motion detectors/sensors, distance sensors, thermometers, light sensors, etc.) and a module (e.g., transceiver, receiver/transmitter) for supporting communication between the aerial mobility devices **600** and/or **610** and the managing device **500** (e.g., throughout the smart building **200**). For example, a sensing module may be equipped with one or more of a camera, a 3D space detection sensor, a distance measuring sensor for measuring a space between the plurality of aerial mobility device **600**, a distance between the aerial mobility device **600** and a take-off/landing area, and the like.

[0035] The taxiway **208** may be a route (or runway) configured for the aerial mobility device **600** to move to a designated area of the vertiport zone **200**. The taxiway **208** may be a mark constructed, for example, as a combination of a line and a color on a floor surface. The designated area may be a specific point of the vertiport zone **200** that is determined based on management information of the aerial mobility devices **600** and **610** and where the managing device **500** lands. The designated area may include at least one of the first take-off/landing area **206**, the object board/alight area **210**, the charging site **214**, the waiting area **216**, and the second take-off/landing area **218**.

[0036] The object board/alight area **210** is a loading/unloading site of a transport object for the aerial mobility devices **600** and **610** and may be a space for a passenger's boarding and alighting and cargo handling. The object board/alight area **210** may be a site where both a passenger and cargo board and alight in a same space, or consist of two sites that are separated for a passenger and cargo respectively.

[0037] The charging site **214** may be equipped with at least one charger, which charges the aerial mobility devices **600** and **610**, and a module that transmits a charge progress state of a device to the managing device **500** and controls charging (e.g., based on information/signals from the management device **500**). The charging site may comprise a charger. As another example, the charging site may comprise various types of energy feeders according to an energy source of the aerial mobility devices **600** and **610**, for example, a feeder that supplies fossil energy-based liquid and/or gas.

[0038] The rotator **212** may be placed in front of the charging site **214** on a movement route to the charging site **214**. If the aerial mobility device **600** enters the rotator **212**, the rotator **212** may rotate the aerial mobility device **600** to make a predetermined portion of the aerial mobility device **600** such as a head directed to the charging site under the control of the managing device **500**.

[0039] The waiting area **216** may be a site where the aerial mobility device **600**, which is going to take off in a predetermined time, waits to be moved to the object board/alight area **210**. The storage area **220** may be a zone where the ground mobility devices **620** and **630** stop and wait for a request of the managing device **500** in the vertiport zone **200**. For example, the ground mobility devices **620** and **630** may be the robot and the purpose-based carrier, and the storage area **220** may not be provided as a site where the aerial mobility device **600** stops or waits. The shift area **222** may be an area in which the aerial mobility device **600** stops over while moving in the complex zone **300** or an area in which the aerial mobility device **600** transferred from the complex zone **300** to the vertiport zone **200** stops over while moving to a designated area.

[0040] The aerial mobility device **600**, which is transferred to the shift area **222**, may be equipped with at least one elevator **224** in order to move to the complex zone **300**. Also, or alternatively, If the aerial mobility device **600** has loaded cargo in the complex zone **300** and is going to take off

with passengers aboard, the elevator **224** may load the aerial mobility device **600**, which is going to take off, and transfer the aerial mobility device **600** to the vertiport zone **200** under the control of the managing device **500**. The elevator **224** may be equipped with a weight measuring module (e.g., weight measurement device comprising a weight measurement sensor, such as a scale) that provides a weight of cargo loaded in the aerial mobility device **600**, which is going to take off, and a weight distribution.

[0041] The complex zone **300** may be used to perform a specific task for the aerial mobility device **600** at a request or control of the managing device **500**. For example, the specific task may be cargo handling, maintenance of the aerial mobility device **600**, charging, and passenger boarding. The complex zone **300** may be equipped with various equipment (e.g., device(s)) for processing a specific task. For example, the equipment may include the robot **620** for handling, maintenance and charging, the carrier **630** for supporting handling and passenger boarding, and service equipment for precise repair. The complex zone **300** may have various modules installed for identifying, transmitting/sending and/or receiving a request and/or information about a situation associated with a specific task between the aerial mobility device **600** and the managing device **500**.

[0042] Hereinafter, referring to FIG. 3, a managing device and a mobility device operated in a smart building will be described according to an example of the present disclosure. FIG. 3 is a view showing constituent modules of a managing device according to an example of the present disclosure.

[0043] The managing device **500** may exchange data with modules (e.g., receivers/transmitters/transceivers) installed in every zone of the smart building **100**, the mobility devices **600** to **630** present in the building, another mobility device linked to the outside thereof, and an external mobility service server, thereby controlling overall operation of the mobility devices inside the building and managing a mobility service provided in the building. Also, or alternatively, the managing device **500** may provide information associated with a mobility service provided in the smart building **100** to a user terminal held by a visitor using the smart building **100** and a user of a mobility device. Furthermore, the managing device **500** may manage various convenience facilities, which stay in the smart building **100** to provide a visitor's convenience, and exchange data with a computing device installed in a convenience facility. As exemplified in FIG. 1, the managing device **500** may be installed inside the smart building **100**, or as another example, may be constructed outside the smart building **100** for remote management.

[0044] The managing device **500** may include a communication unit **510**, an input/out interface **520**, a memory **530**, and a processor **540**.

[0045] The communication unit **510** may communicate with the aerial mobility device **600** using the smart building **100**, the ground mobility devices **620** and **630** supporting the aerial mobility device **600**, another mobility device that is externally linked, sensing modules (e.g., sensors) and control modules (e.g., controllers) constructed in each of the zones **200**, **300** and **400** for controlling the operation and state of a mobility device, and/or a computing module of a convenience facility. The communication unit **510** may receive management information from the aerial mobility device **600**, which is going to land or take off, or an external server controlling the aerial mobility device **600** and transmit an instruction of the device that is generated based on the management information. Also, or alternatively, the communication unit **510** may transmit support-related instruction information, which is generated based on the management information, to the ground mobility devices **620** and **630** and receive data associated with a support state and control from the ground mobility devices **620** and **630**.

[0046] The input/output interface **520** may be equipped with an input device for receiving a request of a manager of the smart building **100**, an output device for providing state and/or control results of a mobility device using the smart building **100**, and/or an adapter interface for connection with another device.

[0047] The memory **530** may store an application and/or a variety of data for integrated operation

of the aerial mobility device **600** using the smart building **100**, the ground mobility devices **620** and **630** and another mobility device. The processor **540** may load the application from the memory **530**, and read data from and/or record data to the memory **530**. Also, or alternatively, the memory **530** may possess internal data used for managing various facilities of the smart building **100** and external data that is transmitted to the building **100**.

[0048] The processor **540** may perform overall control for a mobility device using the smart building **100** and the managing device **100** that operates each of the zones **200** to **400** and facilities. The processor **540** may be configured to execute an application and an instruction stored in the memory **530**. The processor **540** may receive various data from a mobility device using the smart building **100** or supporting a service of the building, each of the zones **200** to **400**, and a convenience facility, create and transmit supervision information, and control the above-listed devices and a target facility by using the information. The supervision information may include data for processing a cooperative operation between a mobility device and each zone.

[0049] The processor **540** may establish (e.g., generate, determine) movement route information of the aerial mobility device **600** in the vertiport zone **200** based on management information associated with managing at least one of a transport object and flight of the aerial mobility device **600** landing on the vertiport zone **200** by using an application, an instruction, and data stored in the memory **530**.

[0050] The management information may include state information, operation information, and flight approval information. As described below, the state information, the operation information, and the flight approval information may each include multiple pieces of detailed information of the management information.

[0051] The state information may include states associated with the operation, control, component and energy refilling of the aerial mobility device **600**. In case the aerial mobility **600** is driven based on an electrical battery, the state information may include, for example, state-of-charge information and aerial vehicle state information. For example, the aerial vehicle state information may include an operation state of a hardware component like a part and an operation state of a software component for flight control. Also, or alternatively, the aerial vehicle state information may include a maintenance request of an abnormal component at an automatic inspection of the aerial mobility device **600** or a pilot's request and an overall weight including the vehicle body, occupants and cargo. In case the aerial mobility device **600** is driven based on a fuel cell or fossil energy, the state information may further include information associated with fuel and a refueling state.

[0052] The operation information may include information associated with managing the flight and transport objects in the aerial mobility device **600**. The flight information may include route information including a depart point, a layover point and a destination, a waiting time at the vertiport zone **200** of the aerial mobility device **600**, and take-off and landing times according to a route of the aerial mobility device **600**. In relation to a transport object, the operation information may include boarding information, cargo information, handling information, and passenger boarding state information. The boarding information may include a current situation of passengers who board or alight from the aerial mobility device **600**. For example, the cargo information may include the weight, size and unloading region of cargo that is loaded in the aerial mobility device **600**. For example, the handling information may be information associated with an individual piece of cargo, which is loaded or unloaded, and a handling region of the cargo. For example, the passenger boarding state information may include an overall weight of passengers boarding the aerial mobility device **600** and in-cabin placement information of the passengers.

[0053] The flight approval information may include permission associated with take-off/landing and flight of the aerial mobility device **600** managed in the vertiport zone **200**, passenger boarding permission, and cargo loading permission.

[0054] If movement route information is established to move to at least one designated area of the

vertiport zone and satisfies a charging permission condition, the processor **540** may set the movement route information to stop at the charging site **214** of the vertiport zone **200**. The processor **540** may control the aerial mobility device **600** to move to the charging site **214** according to the set movement route information and to be charged there. The charging permission condition may be a case where an energy state of the landing aerial mobility device **600** is equal to or less than a threshold value, and the threshold value may be randomly designated or be an energy state that enable flight to a next arrival to be possible.

[0055] Also, or alternatively, the movement route information is established to move the aerial mobility device **600** from the vertiport zone **200** to the complex zone **300** without layover at a designated area associated with take-off/landing, loading/unloading of a transport object, waiting and maintenance, the processor **540** may control the aerial mobility device **600** to move to an exit area of the vertiport zone **200** that approaches the complex zone **300**.

[0056] Also, or alternatively, in order to process cargo handling and maintenance in a predetermined zone, based on management information, the processor **540** may control the aerial mobility device **600**, the ground mobility devices **620** and **630**, and a module for states and controls associated with handling and maintenance. The processor **540** may identify a weight distribution of passengers and/or cargo, provide an optimized position to the aerial mobility device **600** to optimize a location of a transport object inside the aerial mobility device **600** and/or control movement of cargo by the ground mobility devices **620** and/or **630**.

[0057] The processor **540** may be implemented as a single processing module (e.g., a processor) and/or as distributed processing by a plurality of processing modules (e.g., processors). The processor **540** in the present disclosure may collectively refer to the plurality of processing modules.

[0058] FIG. **4** is a view showing constituent modules of a mobility device according to the present disclosure. As described above, a mobility device using the smart building **100** and/or staying in the smart building **100** may include the aerial mobility devices **600** and the ground mobility devices **620** and/or **630**. The aerial mobility devices **600** and/or **610** may be an AAM device and the unmanned aerial vehicle (e.g., a drone). The ground mobility devices **620** and **630** may include the robot **620** and the purpose-based carrier **630**. In FIG. **4**, the aerial mobility device **600** is discussed for clarity of explanation, but FIG. **4** describes a module commonly applicable to any of aerial or ground mobility devices **600**, **610**, **620**, or **640**.

[0059] The mobility device **600** may be moved by manual operation, autonomic control, or a combination thereof. Manual operation may be implemented by a driver or an operator through an interface such as a control device provided in the mobility device **600** or may be implemented by remote control from a control center or an outside control station. Autonomic control, that is, autonomous movement, may be performed by independent processing of the mobility device **600** or may be performed by a combination of remote control through a control center and cooperative processing between the mobility device **600** and the control center.

[0060] The mobility device **600**, which is operated in various forms, may be designed differently according to a uses, a movement space, a driving method, and a control method, but have common functional modules exemplified in FIG. **4** from a comprehensive perspective like mobility. FIG. **4** depicts main common functions in various types of the mobility devices **600**. Accordingly, although an intrinsic functional module used in each type is omitted, an example of the present disclosure does not exclude any module omitted by the mobility device **600** in FIG. **4**, and such a module is not excluded from the scope of right according to this example.

[0061] The mobility device **600** may include a sensor unit **602**, a transceiver **604**, and a load device **606**.

[0062] The sensor unit **602** may be equipped with various types of detectors for sensing various states and situations occurring in external and internal environments of the mobility device **600** and for identifying location information of the mobility device **600**. That is, the sensor unit **602** may

include heterogeneous sensors and obtain sensing data detected from each sensor. The sensor unit **602** may obtain sensor data used for movement control, state data for detecting a state of a module of the mobility device **600**, and situation data for sensing/determining/characterizing a situation of a passenger and/or a load and provide these data to a controller **616** causing a predetermined function and an operation. In the present disclosure, the movement control may be at least one of a linear movement, a turn, and acceleration/deceleration of the mobility device **600**, attitude control of the mobility device **600**, and an operation associated with braking and hovering. The above-described data of the sensor unit **602** is illustrative and further include sensor data for sensing various situations not listed herein.

[0063] The transceiver **604** may support mutual communication with another device outside to exchange data with the other device. For example, the other device may be the managing device **500**, which controls the mobility device or mutually exchanges data associated with movement control of the mobility device, an additional device for supporting movement, another mobility device, and the like. A protocol applied to the transceiver **604** is determined according to a type of the mobility device **600**, and for example, the mobility device **600** may communicate with another vehicle or another device based on cellular communication, wireless access in vehicular environment (WAVE) communication, dedicated short range communication (DSRC) or short range communication, or any other communication scheme. The above-listed communication protocols and schemes are illustrative, and the present disclosure is not limited thereto.

[0064] The load device **606** may be an auxiliary equipment mounted on the mobility device **600**, which consumes electric power supplied from a power source unit **608** by an instruction required for the user of a user or the management of a load or converted from output of the power source unit **608**. The load device **606** may be a type of electric device for non-mobility purpose excluding a mobility power system used in a drive unit **612** in the present disclosure. For example, the load device **606** may be various devices installed in a display system, an air-conditioning system, a light system, a seat system, and the mobility device **600**.

[0065] Although not illustrated in FIG. 4, the mobility device **600** may include an interface for movement control and for receiving a request for an operation of the load device **606**. The interface may be implemented as a hardware device or a software interface. For example, the software interface may be a display with touch input function but is not limited thereto.

[0066] Also, or alternatively, the mobility device **600** may include the power source unit **608**, an operating unit **610**, and the drive unit **612**.

[0067] The power source unit **608** may generate and supply power and electricity used for a mobility power system like the drive unit **612** and the load device **606**. The mobility device **600** may generate energy by at least one of various energy sources. In case the mobility device **600** is driven based on electric energy, for example, the power source unit **608** may be configured as an electric battery or be configured as a combination of an electric battery and a charging module for charging the battery. In case the power source unit **608** is configured as an electric battery alone, the electric battery may be charged from a charging station or another mobility device and supply electricity. In case the power source unit **608** is a combination of an electric battery and a charging module, the power source unit **608** may employ at least one of a fuel cell and a fossil energy-based engine for the charging module. A fuel cell may use a material used for electric power production, for example, hydrogen gas. In case the charging module is an engine, the power source unit **608** may be equipped with a generator coupled with the engine, and the generator may convert mechanical energy generated from the engine to electric energy and charge an electric battery with the converted electric energy.

[0068] As another example, in case the mobility device **600** is driven based on fossil energy, the power source unit **608** may be configured as an internal combustion engine or a turbine engine. As yet another example, the mobility device **600** may have the power source unit **608** which is configured as a hybrid type of a fossil energy-based engine and an electric battery.

[0069] The operating unit **610** and the drive unit **612** may constitute an actuating unit which transfers power generated from the power source unit **608** and externally implements a predetermined moving motion. In the present disclosure, the actuating unit is referred to as an actuator, and these terms may be used interchangeably.

[0070] The operating unit **610** may be equipped with at least one module for implementing a moving operation. In case the mobility device **600** is an aerial mobility device, the operating unit **610** may be equipped with mechanical and software components which perform at least one operation of control of aircraft attitudes such as roll, yaw and/or pitch of the aerial mobility device **30**, hovering control associated with take-off and landing, and flap control for a change of altitude and a turning operation. In case the mobility device **600** is a ground mobility, the operating unit **610** may have mechanical and software components which implement at least one driving operation of longitudinal control such as acceleration and deceleration and transverse control such as steering. The operating unit **610** of the ground mobility may further include a module for transferring power from the power source unit **608** to the drive unit **612** and/or a module for converting power to the drive unit **612** into a predetermined size and a predetermined form.

[0071] The drive unit **612** is a module for externally implementing a linear movement, a turn, and acceleration/deceleration of the mobility device **600**, attitude control of the mobility device **600**, and operations of braking and hovering and may be implemented in various forms according to a type of the mobility device **600**. In the case of the aerial mobility device **600** (e.g., the AAM device), the drive unit **612** may be a rotor-type propeller, flap, and the like, which is installed in at least one of a main wing, a tail wing, and the body. According to specifications of the AAM device **600**, the drive unit **612** may be equipped with a wheel such as a landing gear which is held within a body during flight and is drawn out during take-off and landing. In the case of the small UAV, the drive unit **612** may be composed of a plurality of propellers installed on a frame on an upper body part. In case the mobility devices are the ground mobility devices **620** and **630**, the drive unit **612** may include a plurality of wheels, a driving force transfer module (e.g., engine, motor) for generating and giving a driving force to wheels or for transferring a driving force, a braking module (e.g., brakes) for decelerating the driving of wheels, and a steering module (e.g., steering) for realizing transverse control of wheels. A wheel, a driving force transfer module, and a braking module may constitute a drive assembly, and a plurality of drive assemblies may be provided according to a number of wheels.

[0072] Also, or alternatively, the aerial mobility device **600** may include a storage unit **614** and a controller **616**.

[0073] The storage unit **614** may store an application for controlling the mobility device **600** and various data and load the application or read and record data at a request of the controller **616**. An application and data may be different according to a type and detailed specifications of the mobility device **600** and include sensor data associated with movement control, state data of the mobility associated with movement control, data received from another device, and data associated with energy control between the power source unit **608** and the drive unit **612**. Also, or alternatively to this, the application and the data may include data associated with control of a module in charge of functions other than the above-described controls, software associated with operation of a computing system of the mobility, information and an application for autonomous movement, route information, various information for boarding convenience, and a control program.

[0074] In the present disclosure, the controller **616** may process movement control, route control, energy control, control of the load device **606**, autonomous movement control, and convenience function control by using an application, an instruction, and data which are stored in the storage unit **614**. The controller **616** may also have different control processes according to a type and detailed specifications of the mobility device **600**.

[0075] As an example, the controller **616** may be implemented as a single processing module. As another example, processing according to the above description may be performed as distributed

processing in a plurality of processing modules, and the controller **616** in the present disclosure may collectively refer to the plurality of processing modules.

[0076] Hereinafter, integrated operation by the managing device **500** for the aerial and ground mobility devices **600**, **610**, **620** and **630** using or staying in the smart building **100** and each of the zones **200** to **400** will be described in detail through FIG. 1 to FIG. 7.

[0077] FIG. 5 is a flowchart showing a method for managing a smart building for integrated operation of heterogeneous mobility devices according to another example of the present disclosure. A process according to FIG. 5 to FIG. 7 is performed by the processor **540** of the managing device **500**, but hereinafter, for convenience of explanation, the managing device **500** and the processor **540** may be described interchangeably. Furthermore, in the description below, the aerial mobility device **600** may be described mainly as the AAM device. However, an aerial mobility device may also, or alternatively, include the UAV (e.g., a drone), and the description below may be applicable to the aerial mobility device **610**.

[0078] The managing device **500** may perceive the aerial mobility device **600** approaching the vertiport zone **200** and approve of the aerial mobility device **600** landing, and the aerial mobility device **600** may be controlled by the managing device **500** to land on the take-off/landing areas **206** and **218** (S105).

[0079] The managing device **500** may establish movement route information of the aerial mobility device **600** in the vertiport zone **200**, based on management information associated with at least one of a transport object and flight of the aerial mobility device **600** that is landing (S110).

[0080] As an example, the management information may be transmitted to the managing device **500** via communication with the aerial mobility device **600**. As another example, the management information may be transmitted from an external server controlling the aerial mobility device **600** to the managing device **500**. Also, or alternatively, a part of detailed information constituting the management information may be created by the managing device **500**. For example, handling information and passenger state information, which are collected in the managing device **500** in real time, may be produced by the managing device **500** and be incorporated into the management information.

[0081] For example, in case flight information of the management information is scheduled to load a transport object within a predetermined time and fly to a next destination, movement route information may be created as a route including, as designated areas, the take-off/landing areas **206** and **218**, the object board/alight area **210**, and the waiting area **216**. In case handling information of the management information is indicated to unload loaded cargo in a designated cargo handling area of the object board/alight area **210** or the vertiport zone **200**, the movement route information may be created as a route including, as designated areas, the take-off/landing areas **206** and **218**, the object board/alight area **210** or a cargo handling area and the take-off/landing areas **206** and **218**. In case aerial vehicle state information of the management information is indicated to request inspection and repair for a specific component, the movement route information may be created as a route including, as designated areas, the take-off/landing areas **206** and **218**, the charging site **214** or a designated service area, the waiting area **216** and the take-off/landing areas **206** and **218**. In case flight information is indicated to take off in a predetermined time or aerial vehicle state information is indicated to require precise inspection and in-depth service, the movement route information may be created as a route where the aerial mobility device **600** is moved to an exit area of the vertiport zone **200** approaching the complex zone **300** without layover in any other area of the vertiport zone **200**. For example, the exit area may be the shift area **222** which is a stopping area before moving to an elevator **224**.

[0082] If the movement route information is established to move only to at least one designated area of the vertiport zone **200** (Y of S115), the managing device **500** may set the movement route information to stop at the charging site **214** of the vertiport zone **200**, in response to a charging permission condition being satisfied. Also, or alternatively, the managing device **500** may control

the aerial mobility device **600** to move to the charging site **214** according to the set movement route information (**S120**).

[0083] The managing device **500** may check whether or not the movement route information consists of a designated area of the vertiport zone **200** without the complex zone **300**. The designated area may be an area in the vertiport zone **200** exemplified at step **S110**. After the checking, in case the movement route information consists only of the designated area of the vertiport zone **200**, the managing device **500** may check a charge state of the aerial mobility device **600** based on charge state information and identify whether or not the charge state is equal to or less than a threshold value according to the charging permission condition. For example, the charging permission condition may be randomly designated or be a charged amount required for flight to a next destination. In case the charge state is equal to or less than the threshold value, the charging permission condition is satisfied, and the managing device **500** may modify the pre-established movement route information to stop at the charging site **214**. The movement route information may be modified to arrange a designated area and the charging site in sequence or may be modified in consideration of movement routes of another area mobility **600** and the ground mobility devices **620** and **630**.

[0084] If the aerial mobility device **600** moves to a designated area and the charging site **214** according to the movement route information, the aerial mobility device **600** may move along the taxiway **208** that guides to the area and the charging site **214**. By switching control right to the managing device **500**, the aerial mobility device **600** may be moved by being controlled by the managing device **500**. As another example, the aerial mobility device **600** may be moved by the robot **620** or a cart provided in the vertiport zone **200**. As still another example, as the managing device **500** may transmit the movement route information to the aerial mobility device **600**, the aerial mobility device **600** may be moved by a pilot's taxing manipulation so that movement of the aerial mobility device **600** may be controlled by the managing device.

[0085] As shown in FIG. 2, in case the managing device **500** perceives the aerial mobility device **600** entering the rotator **212** located on a route to the charging site **214**, the managing device **500** may control the rotator **212** to rotate the aerial mobility device **600** so that a predetermined portion such as a head part of the aerial mobility device **600** heads to the charging site **214**. Collision with another mobility device may be prevented which could occur If the aerial mobility device **600** turns in the narrow vertiport zone **200** and then moves to the charging site **214**.

[0086] The managing device **500** may receive charge state information and aerial vehicle state information from the aerial mobility device **600** that is moving to the charging site **214** (**S125**).

[0087] For example, the charge state information may be a remaining battery level, and the aerial vehicle state information may include an operating state of hardware and software components of the aerial mobility device **600** and a maintenance request of an abnormal component. Maintenance by the robot **620** in the charging site **214** may be simple inspection and simple service, and in case a maintenance request is simple inspection and simple service, the managing device **500** may request the robot **620**, which stands by in the storage area **220** or has completed a task, to move to the charging site **214**. In case a maintenance request is precise inspection and in-depth maintenance, the managing device **500** may modify the movement route information again so that the movement route information with layover in the charging site **214** includes the complex zone. The movement route information thus modified again may maintain or exclude the charging site. In case the charging site is maintained, the aerial mobility device **600** may be moved to the complex zone **300** after completion of charging (e.g., based on information indicating the aerial mobility device **600** is charged to a sufficient/threshold level, and the maintenance may be performed in the complex zone **300**. In case the charging site **214** is excluded from the movement route information, the managing device **500** may control the mobility device and each of the zones **200** and **300** in order to process the charging and maintenance of the aerial mobility device **600** in the complex zone **300**.

[0088] As described above, step S125 may be implemented while the aerial mobility device **600** moves to the charging site **214**, but as another example, if the movement route information is completely modified which corresponds to a process implemented before movement of the device, the managing device **500** may receive charge state information and aerial vehicle state information. [0089] The managing device **500** may control the aerial mobility device **600** to be charged in the charging site **214** based on a charge state (S130).

[0090] The charging site **214** may be equipped with a fixed charger, or the robot **620** supporting charging may approach the charging site **214**. The aerial mobility device **600** may be charged by a charger or the robot **620**.

[0091] Also, or alternatively, in case the aerial vehicle state information requests maintenance indicating simple inspection and simple service, the managing device **500** may request the robot **620** supporting maintenance and control maintenance of the aerial mobility device **600**. In this case, the robot **620** and the aerial mobility device **600** may transmit a current state of inspection/service and a result to the managing device **500**, and the managing device **500** may determine whether or not the aerial mobility device **600** is normally operated.

[0092] If the movement route information is established to move from the vertiport zone **200** to the complex zone **300** (N of S115), the managing device **500** may control the aerial mobility device **600** to move to an exit zone of the vertiport zone **200** to approach the complex zone **300**, for example, the shift area **222** (S135).

[0093] The aerial mobility device **600** is moved along the taxiway **208** heading to the shift area **222**, and detailed movement control is actually the same as at step S120. Movement from the shift area **222** to the complex zone **300** may be performed by the elevator **224**, and an operation of the elevator **224** and movement of the aerial mobility device **600** in the complex zone **300** may be controlled by the managing device **500**.

[0094] Even if movement route information is established as a route to the complex zone, If a charging permission condition is satisfied, the managing device **500** may modify the movement route information to include layover at the charging site **214**. The charging permission condition may be the same as described at step S120. As another example, apart from step S120, the charging permission condition may be determined in consideration of flight information of the aerial mobility device **600** and a charging support state of the complex zone **300**. For example, if it is not possible to charge the aerial mobility device **600** for one hour in the complex zone **300** (e.g., if the aerial mobility device **600** is going to take off in one hour), the managing device **500** may determine that the charging permission condition in the charging site **214** is satisfied.

[0095] In case movement route information includes the charging site **214** and the complex zone **300**, the managing device **500** may control the aerial mobility device **600** to move to and be charged in the charging site **214**, which is similar to steps S120 to S130.

[0096] FIG. 6 is a flowchart showing a method for managing a smart building according to still another example of the present disclosure. FIG. 6 is an example that is related to handling and maintenance of the aerial mobility device **600** in the complex zone **300** and an arrangement setting of a transport object for safe flight.

[0097] The managing device **500** may perceive the aerial mobility device **600** entering the complex zone **300** by the elevator **224**, check handling information of management information, and control the aerial mobility device **600**, the robot **620** of the complex zone **300**, and a relevant module to load or unload cargo based on the handling information (S205).

[0098] The aerial mobility device **600** may be moved to a preset area of the complex zone **300**, and the movement may be performed similar to what is described at step S120. The managing device **500** may check handling information associated with cargo to be loaded in or unloaded from the aerial mobility device **600** and also check a weight resulting from cargo to be loaded, an unloading region of a planned flight route, and a passenger boarding state of planned flight. Based on the above-described information and state, the managing device **500** may create cargo arrangement

information with a loaded form of cargo inside the aerial mobility device **600**. The managing device **500** may control the robot **620** provided in the complex zone **300** and load cargo in a designated location in the aerial mobility device **600** according to cargo arrangement information. As another example, the managing device **500** may control the robot **620**, which is parked in the storage area **220** of the vertiport zone **200** or has completed a task, to move to the complex zone **300** and transmit detailed information such as instruction information according to the above-described state and information to the robot **620**. The managing device **500** may control the robot **620** to perform processing according to instruction information, for example, unloading cargo according to handling information in the aerial mobility device **600**. As another example, cargo arrangement information may be transmitted to a device (e.g., carrier and/or robot) of a manager that loads baggage so that the manager may load cargo in a designated location, and thus the managing device **500** may control cargo loading.

[0099] During a handling task or after completion thereof, the managing device **500** may control the aerial mobility device **600**, the robot **620** of the complex zone **300**, a charger and a relevant module to perform at least one of maintenance and charging of the aerial mobility device **600** based on state information including charge state information and aerial vehicle state information (**S210**).

[0100] For example, maintenance may be precise inspection and in-depth service, and as an example, for overall maintenance, simple inspection and simple service may be additionally included. For example, in case movement is performed to the complex zone **300** without charging in the charging site **214**, charging in the complex zone **300** may be performed based on a charged amount that is determined according to flight information and cargo information. At least one of maintenance and charging may be performed in an area, where a handling task is processed, or another area. In case neither charging nor maintenance is required based on charge state information and aerial vehicle state information, step **S210** may be omitted.

[0101] The managing device **500** may control the aerial mobility device **600** and the elevator **224** to move the aerial mobility device **600** to the vertiport zone **200** by using the elevator **224** (**S215**).

[0102] The managing device **500** may measure a weight resulted from loaded cargo and a weight distribution by using a weight measuring module (e.g., weight measurement device comprising a weight measurement sensor, such as a scale) mounted in the elevator **224** and determine whether the weight distribution satisfies a weight balance (**S220**).

[0103] A weight distribution may be generated to reflect an impact from a plurality of unloading regions according to a flight route and a passenger boarding state associated with an arrangement of seated passengers in the aerial mobility device **600** that is going to take off. Specifically, even if cargo is loaded in a designated location of the aerial mobility device **600** according to cargo arrangement information at step **S205**, if the aerial mobility device **600** is moved to the vertiport zone **200** after a long time, a final boarding passenger, final loaded cargo, and an ultimately determined layover point may be modified. Accordingly, the managing device **500** may reflect an impact of an unloading region and an arrangement of seated passengers according to modifications in a weight distribution. If the aerial mobility device **600**, which is about to take off, is moved by the elevator **224**. As another example, in case no cargo arrangement information is created at step **S205** and cargo is randomly loaded in the aerial mobility device **600**, an unloading region and an arrangement of seated passengers of the aerial mobility device **600**, which is about to take off, are considered through step **S220**, and thus a weight balance may be identified.

[0104] For example, a weight balance may be a state of satisfying a reference safety degree required for the body and cabin of the aerial mobility device **600** in consideration of a flying body shape, a body center according to a weight distribution of transport objects, flight control required according to an aerial vehicle state and a surrounding environment, and flight safety.

[0105] If a weight distribution does not satisfy a weight balance, the managing device **500** may create arrangement information including at least one of cargo arrangement information and passenger arrangement information in order to strike the weight balance (**S225**).

[0106] In a designated area of the vertiport zone **200**, the managing device **500** may control to place at least one of cargo and a passenger in a designated location in the aerial mobility device **600** based on the arrangement information (**S230**).

[0107] The designated area is an area of the vertiport zone **200**, which is set according to movement route information, and may be at least one of, for example, the object board/alight area **210**, the waiting area **216**, and a cargo loading area.

[0108] The managing device **500** may control the robot **620** provided in the vertiport zone **200** and load cargo in a modified location in the aerial mobility device **600** according to cargo arrangement information. The robot **620** may be a robot that is parked in the storage area **220** or has completed a task. As another example, cargo arrangement information may be transmitted to a device of a manager that loads baggage so that the manager may load cargo in a modified location, and thus the managing device **500** may control cargo loading.

[0109] As the managing device **500** provides passenger arrangement information to a passenger or a manager (e.g., via a device), the managing device **500** may control a seated arrangement of passengers.

[0110] If a weight distribution satisfies a weight balance, the managing device **500** may not modify (e.g., may maintain) arrangement information of cargo and/or passengers of the aerial mobility device **600** moved to the vertiport zone **200** (**S235**). The passengers may be seated in locations that are previously noticed/detected, no change of location may occur to cargo.

[0111] FIG. 7 is a flowchart showing a method for managing a smart building according to yet another example of the present disclosure. FIG. 7 is an example associated with an operation of a ground mobility device that supports a task for the aerial mobility device **600** in the vertiport zone **200**.

[0112] The managing device **500** may control ground mobility devices supporting a task of the aerial mobility device **600**. For example, the managing device **500** may control the robot **620** and/or the carrier **630** to be parked in the storage area **220** (**S305**).

[0113] The robot **620** may move between the vertiport zone **200** and another zone and carry cargo in each zone or between zones. Another zone may be the complex zone **300** or the supplementary zone **400**. The robot **620** may be equipped with an arm for carrying cargo. Apart from carrying cargo, the robot **620** may be equipped with a module for performing at least one of maintenance and/or charging. The robot **620** may transmit a progress, a result and a current location, which are associated with the above-described processing, to the managing device **500**. The carrier **630** may transport a passenger using the aerial mobility device **600** to the vertiport zone **200** or another zone by moving between the vertiport zone **200** and another zone.

[0114] If processing requested by the managing device **500** is completed, the robot **620** and the carrier **630** may be controlled to stay in the storage area **220**, and the storage area **220** may be provided in separation from a designated area of the aerial mobility device **600**.

[0115] The managing device **500** may receive detailed information on at least one of aerial vehicle state information, charge state information, handling information, flight information, and boarding information of the aerial mobility device **600** that is going to land on the vertiport zone **200** (**S310**).

[0116] If the aerial mobility device **500** enters a designated area designated by movement route information, the managing device **500** may control the robot **620** and/or the carrier **630** to make at least one of the robot **620** and the carrier **630** approach the aerial mobility device **600** (**S315**).

[0117] For example, the designated area may be the object board/alight area **210**, the waiting area **216** and the charging site **214**, and in case the aerial mobility device **600** enters a designated area associated with the detailed information, at least one of the robot **620** and the carrier **630** may approach at least one of the robot **620** and the carrier **630** that moves to the designated area.

[0118] Allocation and a movement route of the robot **620** and the carrier **630** supporting the aerial mobility device **600** may be determined based on a degree of congestion caused by the aerial mobility device **600** in the vertiport zone **200**. For example, the degree of congestion may be

created based on a stop situation of the aerial mobility device **600** and **610** in the vertiport zone **200**, turn around time (TAT) thereof, and the like. In case the degree of congestion exceeds a threshold value, the managing device **500** may control, based on the degree of congestion, the robot **620** and/or the carrier **630** to adjust at least one of whether or not to allocate the robot **620** and/or the carrier **630** in the vertiport zone **200**, an approaching time, and a movement route.

[0119] The managing device **500** may transmit instruction information based on detailed information to at least one of the robot **620** and the carrier **630** and control at least one of the robot **620** and the carrier **630** to perform processing, according to the instruction information, in/associated with the aerial mobility device **600** (S320).

[0120] The present disclosure is technically directed to providing a system and method for managing a smart building that maximizes operation processing of an aerial mobility device and also enables the aerial mobility device to establish smooth support and integrated link to a ground mobility device.

[0121] The technical problems solved by the present disclosure are not limited to the above described technical problems. Other technical problems that are not described herein should be more clearly understood by a person having ordinary skill in the technical field, to which the present disclosure belongs, from the following description.

[0122] According to the present disclosure, a system is provided for managing a smart building for integrated operation of heterogeneous mobility device. The system may comprising: a smart building with a vertiport zone and a complex zone for supporting operation of an aerial mobility device and a ground mobility device; and a managing device equipped with a memory configured to store at least one instruction and at least one processor configured to execute the instruction stored in the memory, wherein the processor is further configured to: establish movement route information of the aerial mobility device in the vertiport zone based on management information associated with management of at least one of a transport object and flight of the aerial mobility device that lands on the vertiport zone, in case that the movement route information is established (e.g., generated, determined, received) and indicates to move to at least one designated area of the vertiport zone and a charging permission condition is satisfied, set the movement route information to have a charging site in the vertiport zone as a layover point and control the aerial mobility device to move to the charging site according to the set movement route information and to be charged, and in case that the movement route information is established to move the aerial mobility device to the complex zone, control the aerial mobility device to move to an exit area of the vertiport zone approaching the complex zone.

[0123] According to an example of the system of the present disclosure, controlling of the aerial mobility device to move to the exit area may comprises: in case that the movement route information is established to head to the complex zone and the charging permission condition is satisfied, setting the movement route information to have the charging site as a layover point for charging the aerial mobility device; controlling the aerial mobility device to move to the charging site; and controlling the aerial mobility device to move to the exit area after completion of the charging.

[0124] According to an example of the system of the present disclosure, controlling of the aerial mobility device to be charged may comprises: receiving the management information associated with a charge state and an aerial vehicle state from the aerial mobility device that moves to the charging site; controlling the aerial mobility device to be charged in the charging site based on the charge state; and controlling the ground mobility device to perform maintenance of the aerial mobility device in the charging site based on the aerial vehicle state.

[0125] According to an example of the system of the present disclosure, the vertiport zone further may include a rotator placed in front of the charging site on a movement route to the charging site, and wherein the moving of the aerial mobility device to the charging site comprises, in response to the aerial mobility device entering the rotator, rotating, by the processor, the aerial mobility device

to make a predetermined portion of the aerial mobility device directed to the charging site.

[0126] According to an example of the system of the present disclosure, the complex zone is may a zone where at least one of cargo handling and maintenance is performed on the aerial mobility device, and the management information includes handling information associated with the cargo handling, an aerial vehicle state, and a charge state, and wherein the processor is further may configured to: control loading or unloading of the cargo in the complex zone based on the handling information, and control to perform at least one of maintenance and charging of the aerial mobility device in the complex zone based on the aerial vehicle state and the charge state.

[0127] According to an example of the system of the present disclosure, the vertiport zone and the complex zone are vertically may arranged, wherein the smart building further includes an elevator that transfers the aerial mobility device between the vertiport zone and the complex zone, wherein the elevator is equipped with a weight measuring module that provides a weight resulting from cargo loaded in the aerial mobility device and a weight distribution, and wherein the processor is further may configured to: in case that the weight distribution dose not satisfy a weight balance, create arrangement information including at least one of cargo arrangement information and passenger arrangement information in order to strike the weight balance, and control, in the designated area, to place at least one of cargo and a passenger in a designated location in the aerial mobility device based on the arrangement information.

[0128] According to an example of the system of the present disclosure, the weight distribution is may create to reflect an impact from a plurality of unloading regions according to a flight route and a passenger boarding state associated with an arrangement of seated passengers in the aerial mobility device that is going to take off.

[0129] According to an example of the system of the present disclosure, the ground mobility device may include a carrier and a robot that carry a transport object between the vertiport zone and another zone, and the management information includes an aerial vehicle state, a charge state, handling information associated with cargo handling for the aerial mobility device, flight information, and boarding information of a passenger, and wherein the processor is further may configured to: control the robot and the carrier to be parked in a storage zone, receive detailed information on at least one of the aerial vehicle state, the charge state, the handling information, the flight information, and the boarding information of the aerial mobility device that is going to land on the vertiport zone, in response to the aerial mobility device landing and entering the designated area, control at least one of the robot and the carrier to approach the aerial mobility device, and control at least one of the robot and the carrier to perform processing according to instruction information in the aerial mobility device by transmitting instruction information based on the detailed information to at least one of the robot and the carrier.

[0130] According to an example of the system of the present disclosure, the processor is further may configured to: control the robot to move to the complex zone where the aerial mobility device stops, and control the robot to perform processing according to instruction information in the aerial mobility device by transmitting the instruction information according to the detailed information to the robot.

[0131] According to an example of the system of the present disclosure, in case that a degree of congestion caused by the aerial mobility device in the vertiport zone exceeds a threshold value, the processor is further may configured to control at least one of the robot and the carrier to move in the vertiport zone based on the degree of congestion.

[0132] According to another example of the present disclosure, a method is provided for managing a smart building for integrated operation of heterogeneous mobility devices. The smart building may has a vertiport zone and a complex zone for supporting operation of an aerial mobility device and a ground mobility device, and the method may comprises: establishing movement route information of the aerial mobility device in the vertiport zone based on management information associated with management of at least one of a passenger, cargo, and flight of the aerial mobility

device that lands on the vertiport zone; in case that the movement route information is established to move to at least one designated area of the vertiport zone and a charging permission condition is satisfied, setting the movement route information to have a charging site in the vertiport zone as a layover point and controlling the aerial mobility device to move to the charging site according to the set movement route information and to be charged; and in case that the movement route information is established to move the aerial mobility device to the complex zone, controlling the aerial mobility device to move to an exit area of the vertiport zone approaching the complex zone.

[0133] The features of the present disclosure, which are briefly summarized above, are only examples of aspects or features of the present disclosure and detailed description of the disclosure follows and are not intended to limit the scope of the present disclosure.

[0134] The technical problems solved by the present disclosure are not limited to the above mentioned technical problems. Other technical problems solved by the present disclosure, which are not described herein should be more clearly understood by a person having ordinary skill in the art or technical field to which the present disclosure belongs from the following description.

[0135] According to the present disclosure, it is possible to provide a system and method for managing a smart building that maximizes operation processing of an aerial mobility device and also enables the aerial mobility device to establish smooth support and integrated link to a ground mobility device.

[0136] The effects obtainable from the present disclosure are not limited to the above-mentioned effects. Other effects not mentioned herein should be more clearly understood by those of ordinary skill in the art through the following description.

[0137] For example, the processing may be at least one of passenger board/alight, cargo handling, maintenance, and charging. The instruction information may be created based on detailed information that requires the processing.

[0138] While the methods of the present disclosure described above are represented as a series of operations for clarity of description, it is not intended to limit the order in which the steps are performed. The steps described above may be performed simultaneously or in different order as necessary. In order to implement the method according to the present disclosure, the described steps may further include different or other steps, may include remaining steps except for some of the steps, or may include other additional steps except for some of the steps.

[0139] The various examples of the present disclosure do not disclose a list of all possible combinations and are intended to describe representative aspects of the present disclosure. Aspects or features described in the various examples may be applied independently or in combination of two or more.

[0140] Also, or alternatively, various examples of the present disclosure may be implemented in hardware, firmware, software, or a combination thereof. In the case of implementing the present disclosure by hardware, the present disclosure can be implemented with application specific integrated circuits (ASICs), Digital signal processors (DSPs), digital signal processing devices (DSPDs), programmable logic devices (PLDs), field programmable gate arrays (FPGAs), general processors, controllers, microcontrollers, microprocessors, etc.

[0141] The scope of the disclosure includes software or machine-executable commands (e.g., an operating system, an application, firmware, a program, etc.) for enabling operations according to the methods of various examples to be executed on an apparatus or a computer, a non-transitory computer-readable medium having such software or commands stored thereon and executable on the apparatus or the computer.

Claims

1. A system comprising: a smart building comprising: a vertiport zone, and a complex zone configured to support operation of an aerial mobility device, that lands in the vertiport zone, and a

ground mobility device; and a managing device comprising: at least one processor, and a memory storing at least one instruction that, when executed by the at least one processor, configures the at least one processor to: generate, based on management information, movement route information of the aerial mobility device in the vertiport zone, wherein the management information is associated with management of at least one of: a transport object for transport by the aerial mobility device, or flight of the aerial mobility device, based on the movement route information indicating at least one designated area of the vertiport zone and on a charging permission condition being satisfied, set the movement route information to comprise a layover point of a charging site in the vertiport zone and control the aerial mobility device to move to the charging site according to the set movement route information and to be charged, and based on the movement route information indicating to move the aerial mobility device to the complex zone, control the aerial mobility device to move to an exit area, of the vertiport zone, approaching the complex zone.

2. The system of claim 1, wherein the at least one instruction, when executed by the at least one processor, configures the at least one processor to control of the aerial mobility device to move to the exit area by: setting, based on the charging permission condition being satisfied, the movement route information to comprise the charging site as a layover point for charging the aerial mobility device; controlling the aerial mobility device to move to the charging site; and controlling the aerial mobility device to move to the exit area after completion of the charging.

3. The system of claim 1, wherein the at least one instruction, when executed by the at least one processor, configures the at least one processor to control the aerial mobility device to be charged by: receiving, from the aerial mobility device, the management information associated with a charge state and an aerial vehicle state; controlling, based on the charge state, the aerial mobility device to be charged in the charging site; and controlling, based on the aerial vehicle state, the ground mobility device to perform maintenance of the aerial mobility device in the charging site.

4. The system of claim 1, wherein the vertiport zone further comprises a rotator placed in front of the charging site on a movement route to the charging site, and wherein the at least one instruction, when executed by the at least one processor, configures the at least one processor to control the aerial mobility device to move to the charging site by, based on the aerial mobility device entering the rotator, rotating the aerial mobility device to make a predetermined portion of the aerial mobility device directed to the charging site.

5. The system of claim 1, wherein the complex zone comprises a zone configured for at least one of cargo handling or maintenance of the aerial mobility device, and the management information comprises at least one of: handling information associated with the cargo handling; an aerial vehicle state; or a charge state; and wherein the at least one instruction, when executed by the at least one processor, further configures the at least one processor to: control, based on the handling information, loading or unloading of the cargo in the complex zone, and control, based on the aerial vehicle state and the charge state, performance of at least one of maintenance or charging of the aerial mobility device in the complex zone.

6. The system of claim 1, wherein the vertiport zone and the complex zone are vertically arranged; wherein the smart building further comprises an elevator configured to transfer the aerial mobility device between the vertiport zone and the complex zone; wherein the elevator is equipped with a weight measurement device configured to acquire a weight measurement of cargo loaded in the aerial mobility device and a weight distribution associated with the aerial mobility device; and wherein the at least one instruction, when executed by the at least one processor, further configures the at least one processor to: based on the weight distribution not satisfying a weight balance, create arrangement information configured to cause the weight distribution to satisfy the weight balance, wherein the arrangement information comprises at least one of cargo arrangement information or passenger arrangement information; and cause, based on the arrangement information and in the at least one designated area, placement of at least one of cargo or a passenger in a designated location in the aerial mobility device.

- 7.** The system of claim 6, wherein the at least one instruction, when executed by the at least one processor, further configures the at least one processor to create the arrangement information to cause the weight distribution to reflect at least one of: a plurality of unloading regions associated with a flight route of the aerial mobility device, or a passenger boarding state associated with an arrangement of seated passengers in the aerial mobility device.
- 8.** The system of claim 1, wherein the ground mobility device comprises a robot and a carrier configured to carry a transport object between the vertiport zone and another zone, and the management information comprises at least one of an aerial vehicle state, a charge state, handling information associated with cargo handling for the aerial mobility device, flight information, or boarding information of a passenger; and wherein the at least one instruction, when executed by the at least one processor, further configures the at least one processor to: control the robot and the carrier to be parked in a storage zone, receive information about at least one of the aerial vehicle state, the charge state, the handling information, the flight information, or the boarding information of the aerial mobility device, based on the aerial mobility device entering the designated area, control at least one of the robot or the carrier to approach the aerial mobility device, and sending an instruction, based on the received information and to at least one of the robot or the carrier, configured to control the at least one of the robot or the carrier to perform processing associated with the received information.
- 9.** The system of claim 8, wherein the at least one instruction, when executed by the at least one processor, further configures the at least one processor to: control the robot to move to the complex zone comprising the aerial mobility device; and send, to the robot, the instruction configured to control the robot to perform the processing.
- 10.** The system of claim 8, wherein the at least one instruction, when executed by the at least one processor, further configures the at least one processor to, based on a degree of congestion caused by the aerial mobility device in the vertiport zone exceeding a threshold value, control at least one of the robot or the carrier to move, based on the degree of congestion, in the vertiport zone.
- 11.** A method for managing a smart building comprising: a vertiport zone, and a complex zone configured to support operation of an aerial mobility device and a ground mobility device, wherein the method comprises: generating, based on management information, movement route information of the aerial mobility device in the vertiport zone, wherein the management information is associated with management of at least one of: a transport object for transport by the aerial mobility device, or flight of the aerial mobility device; based on the movement route information indicating at least one designated area of the vertiport zone and on a charging permission condition being satisfied, setting the movement route information to comprise a layover point of a charging site in the vertiport zone and controlling the aerial mobility device to move to the charging site according to the set movement route information and to be charged; and based on the movement route information indicating to move the aerial mobility device to the complex zone, controlling the aerial mobility device to move to an exit area, of the vertiport zone, approaching the complex zone.
- 12.** The method of claim 11, wherein the controlling of the aerial mobility device to move to the exit area comprises: setting, based on the charging permission condition being satisfied, the movement route information to comprise the charging site as a layover point for charging the aerial mobility device; controlling the aerial mobility device to move to the second charging site; and controlling the aerial mobility device to move to the exit area after completion of the charging.
- 13.** The method of claim 11, wherein the controlling of the aerial mobility device to be charged comprises: receiving, from the aerial mobility device, the management information associated with a charge state and an aerial vehicle state; controlling, based on the charge state, the aerial mobility device to be charged in the charging site; and controlling, based on the aerial vehicle state, the ground mobility device to perform maintenance of the aerial mobility device in the charging site.
- 14.** The method of claim 11, wherein the vertiport zone further comprises a rotator placed in front of the charging site on a movement route to the charging site, and wherein the controlling the aerial

mobility device to move to the charging site comprises, rotating, based on the aerial mobility device entering the rotator, the aerial mobility device to make a predetermined portion of the aerial mobility device directed to the charging site.

15. The method of claim 11, wherein the complex zone comprises a zone configured for at least one of cargo handling and maintenance of the aerial mobility device, and the management information comprises at least one of: handling information associated with the cargo handling; an aerial vehicle state; or a charge state; and wherein the method further comprises: after the controlling of the aerial mobility device to move to the exit area, controlling, based on the handling information, loading or unloading of the cargo in the complex zone; and controlling, based on at least one of the aerial vehicle state or the charge state, performance of at least one of maintenance or charging of the aerial mobility device in the complex zone.

16. The method of claim 11, wherein the vertiport zone and the complex zone are vertically arranged, wherein the smart building further comprises an elevator configured to transfer the aerial mobility device between the vertiport zone and the complex zone, wherein the elevator is equipped with a weight measurement device configured to acquire a weight measurement of cargo loaded in the aerial mobility device and a weight distribution associated with the aerial mobility device, and wherein the method further comprises: based on the weight distribution not satisfying a weight balance, creating arrangement information configured to cause the weight distribution to satisfy the weight balance, wherein the arrangement information comprises at least one of cargo arrangement information or passenger arrangement information; and cause, based on the arrangement information, placement of at least one of cargo or a passenger in a designated location in the aerial mobility device.

17. The method of claim 16, wherein the creating the arrangement information causes the weight distribution to reflect at least one of: a plurality of unloading regions associated with a flight route of the aerial mobility device, or a passenger boarding state associated with an arrangement of seated passengers in the aerial mobility device.

18. The method of claim 11, wherein the ground mobility device comprises a robot and a carrier configured to carry a transport object between the vertiport zone and another zone, and the management information comprises at least one of an aerial vehicle state, a charge state, handling information associated with cargo handling for the aerial mobility device, flight information, or boarding information of a passenger; and wherein method further comprises: controlling the robot and the carrier to be parked in a storage zone, receiving information about at least one of the aerial vehicle state, the charge state, the handling information, the flight information, or the boarding information; based on the aerial mobility device entering the designated area, controlling at least one of the robot or the carrier to approach the aerial mobility device; and sending an instruction, based on the received information and to at least one of the robot or the carrier, configured to control the at least one of the robot or the carrier to perform processing associated with the received information.

19. The method of claim 18, further comprising: controlling the robot to move to the complex zone comprising the aerial mobility device; and sending, to the robot, the instruction configured to control the robot to perform the processing.

20. The method of claim 18, further comprising, based on a degree of congestion caused by the aerial mobility device in the vertiport zone exceeding a threshold value, controlling at least one of the robot or the carrier to move, based on the degree of congestion, in the vertiport zone.
